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Overreaction in the Brazilian stock market

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Abstract

The overreaction hypothesis on a new set of data for the Brazilian stock market over the period 1970–1989 is investigated. Both market adjusted returns and the standard Sharpe–Lintner CAPM adjusted returns are used. Price reversals in 2-year returns are detected and the results contrast with the U.S. evidence in that the magnitude of the effect is more pronounced than in the U.S. The results also suggest that differences in risk, as measured by CAPM-betas using the method suggested by Chan (Chan, K.C., 1988, On the contrarian investment strategy, *Journal of Business* 61, 147–163), cannot account for the overreaction effect. I also examine the issue of asymmetry versus symmetry in the overreaction effect. Using the criterion proposed by Dissanaike (Dissanaike, G., 1992, Are stock price reversals really symmetric? University of Cambridge Department of Applied Economics Discussion Paper, AF series, No. 4) I find evidence that the price reversals are asymmetric.

Key words: Overreaction hypothesis; Asymmetry

JEL classification: G14

1. Introduction

With the surge of institutional interest in emerging capital markets, it is important to understand how well these markets operate. Indeed, little attention has been focused on the informational efficiency of capital markets in developing countries. For example,¹ not much is known about the pricing

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¹ A recent exception is the work of Harvey (1993).

of financial assets in Brazil. Some previous research suggests that some of the patterns observed in U.S. stock returns are also present in the Brazilian stock market. For example, studies by Lemgruber et al. (1989) and Costa (1990) detected a weekend seasonality similar to that observed in the U.S. market. Costa and O'Hanlon (1991) found that average returns were related to market capitalization. Interestingly, there is no evidence of monthly seasonality in Brazilian returns.

In recent years much attention has been focused on the apparent tendency of markets to 'over-react' to good and bad news. De Bondt and Thaler (1985, 1987), Howe (1986) and Chopra et al. (1991) argue that the overreaction is a manifestation of a fundamental market inefficiency. Others suggest that the apparent overreaction is due to risk mismeasurement or is just another version of the size effect (see Chan, 1988; Ball and Kothari, 1989; Zarowin, 1990).

Overreaction can be said to occur when individuals, in revising their beliefs, tend to overweight recent information (in violation of Bayes' rule) and underweight prior data, De Bondt and Thaler (1985, p. 793). The methodology used by De Bondt and Thaler to identify this anomaly in the U.S. market involved the construction of two portfolios during a portfolio formation period, one of the portfolios (the 'winner' portfolio) being composed of extreme high return securities and the other portfolio (the 'loser' portfolio) being composed of extreme low return securities. When comparing the performance of these two portfolios in the subsequent test period, they found that the 'loser' portfolio significantly outperformed the 'winner'.

My objective is to test whether a similar effect is found in the Brazilian stock market, using monthly data collected for the São Paulo Stock Exchange (BOVESPA) from January 1970 to December 1989. The organization of the paper is as follows. In the next section, some institutional background on the São Paulo Stock Exchange is provided. The data collection and the methodology are detailed in the third section. In Section 4 the results are presented. Some concluding remarks are offered in the final section.

2. The São Paulo stock exchange

The stock market in Brazil is small by international comparisons. By the end of 1989, corresponding to the last year of our study period, there were 622 companies listed on stock exchanges. Of these, less than half are actively traded. There are nine stock exchanges in Brazil, however, the São Paulo Stock Exchange (BOVESPA) handled 57.6% of the total volume traded on the domestic market in 1989. The market value of the stocks traded in BOVESPA was US\$ 44 billion in December 1989.

2.1. Foreign investment, capital tax and commission rates

During the period of this study, direct investment of foreign capital in the Brazilian exchanges was not allowed. Foreign investment was restricted to investment funds, external debt conversion funds and stock portfolios managed by international financial institutions. Dividends are subject to income tax while capital gains on share trading were not taxed until the beginning of 1990. Commission rates are fixed depending on the value of the transaction. Rates start at 2%, and fall to 0.5% for transactions valued at US\$2,500 or more, with a minimum commission of US\$2.00.

3. Data description and methodology

3.1. Data description

The data used in this study includes the monthly prices of a sample of 121 stocks traded in BOVESPA during the period January 1970 to December 1989. These stocks accounted for more than 70% of the traded volume in that period. In order to minimize my survivorship bias, the sample included all stocks that belonged to the BOVESPA index (IBOVESPA) for at least four years consecutively at any time during the experimental period.² And, when a company is delisted the security final return is set to minus one.

The risk free rate used was the return rate of national treasury bills. These bills offer an interest payment in addition to an adjustment to compensate for price-level changes. As a surrogate for the market index we constructed an equally weighted index of the 121 stocks used in this study.

3.2. Source of information

Most of the data were obtained in the São Paulo Stock Exchange where there is a relatively well-organized file with information about all securities traded. Since 1972 a great amount of information has been recorded on magnetic tapes (closing and opening monthly prices, splits, rights, cash dividends, etc.). Unfortunately, not all of these files are complete. Thus complementary material was used. The publications consulted were *Anuário da Bolsa* (Annual Bulletin of the Stock Exchange) and *Boletim Diário de Informações* (Daily Information Bulletin). For the monthly rate of return on treasury bills the publication consulted was *Revista ANDIMA* (Journal of the National Association of Listed Companies).³

² This index is the official index of the São Paulo Stock Exchange. The weighting criteria is based on the volume of trading, and it is computed based on a basket of stocks that represent 80% of the total trading volume. Its composition is reviewed every four months and it assumes the reinvestment of dividends and sale of rights in the same stock.

³ Further details about data collection and sources can be obtained from the author on request.

3.3. Methodology

The monthly returns of each stock in the sample are continuously compounded returns. They are computed as the difference between the natural logarithms of two consecutive monthly closing prices and are adjusted for dividends, rights, splits, etc.

Both market adjusted returns and the standard Sharpe–Lintner CAPM adjusted returns are used.

As the results were very similar from the use of these two methods, we present only the results using market adjusted returns which are defined:

$$u_{j,t} = R_{j,t} - R_{m,t}, \quad (1)$$

where $u_{j,t}$ is the market adjusted abnormal return of stock j in month t ; and $R_{m,t}$ is the return of the equally-weighted index of Brazilian stock exchange in month t .

3.4. Number of periods used

The test periods spanned 16 years of monthly returns from 1974 to 1989 inclusive. In addition, returns from 1972 and 1973 are used for initial portfolio formation.

The availability of 16 years of monthly returns provide 8 non-overlapping test periods, the first of these (from 1974 to 1975) is based on portfolios formed in 1972–1973 and the last (from 1988 to 1989) is based on portfolios formed in 1986–1987.

3.5. Portfolio formation

For each of the 8 portfolio formation periods (1972/73 to 1986/87) we computed the cumulative market adjusted abnormal return for every stock:

$$CU_j = \sum_{t=1}^T u_{j,t}, \quad (2)$$

where CU_j is the cumulative market adjusted abnormal return of stock j in month t ; and T is the last month ($t=24$) of a formation period.

Then, at month T ($t=24$) the cumulative abnormal returns were ranked from low to high and 5 portfolios were formed. For every test period we

computed the cumulative average abnormal returns (*CAR*) of all stocks in each of the 5 portfolios during the 24 months of the test period:

$$CAR_{i,N,t} = \sum_t \left((1/n) \sum_{j=1}^n u_{j,t} \right) \text{ for } i=1, \dots, 5, \quad (3)$$

where N is the test period in question, $N=1, \dots, 8$; n is the number of stocks assigned to each portfolio, not necessarily the same number for each of the 5 portfolios;⁴ $CAR_{i,N,t}$ is the cumulative average market adjusted abnormal return in month t of the test period N , for each one of the 5 portfolios; $u_{j,t}$ is the abnormal return of stock j in month t .

The above procedure involves an implicit strategy of rebalancing the portfolio each month so that each stock's cumulative abnormal return has equal weight. When a stock is no longer available after the portfolio formation period, the portfolio *CAR* was averaged over the available cumulative abnormal returns.

Next for each test period we calculated the average *CAR* for each of the 5 portfolios in each month ($t=1, \dots, 24$) of the test period.

$$ACAR_{i,t} = \frac{\sum_{N=1}^8 (CAR_{i,N,t})}{8} \text{ for } t=1, \dots, 24 \text{ and } i=1, \dots, 5, \quad (4)$$

where $ACAR_{i,t}$ is the average *CAR* for each of the 5 portfolios in month t ; and N is the number of test periods, $N=1, \dots, 8$.

The *CARs* and *ACARs* were then used to test the performance of each portfolio and to compare the performances of the 'winner' and 'loser' portfolios.

3.6. Statistical tests used

In order to test the overreaction effect in the Brazilian market we tested the following null hypotheses:

$$H_1: E(ACAR_{1,t}) = 0, t=1, \dots, 24;$$

$$H_2: E(ACAR_{5,t}) = 0, t=1, \dots, 24;$$

$$H_3: E(ACAR_{1,t} - ACAR_{5,t}) = 0, t=1, \dots, 24.$$

The first two hypotheses were tested by carrying out a t -test on the mean of $ACAR_{1,t}$ and $ACAR_{5,t}$, with unknown population variance. The test statistics are given by:

$$t_{1,t} = \frac{ACAR_{1,t}}{S_1/\sqrt{N}}, \quad (5)$$

⁴ In 1974 each portfolio contained about 9 stocks and about 18 stocks in 1989.

Table 1

Average cumulative average market adjusted abnormal returns – ACAR – for 8 two-year test periods (1974–1989)

Months after formation period	Portfolios (ACAR%)					
	$ACAR_{1,t}$	$ACAR_{2,t}$	$ACAR_{3,t}$	$ACAR_{4,t}$	$ACAR_{5,t}$	$ACAR_{1,t} - ACAR_{5,t}$
6	6.33 (1.28)	-0.56 (-0.12)	0.24 (0.06)	-6.24 (-1.45)	-4.48 (-1.07)	10.81 (1.67) ($H = 1.59$)
12	11.82 (1.71)	12.25 (2.56)*	-2.66 (-0.51)	0.05 (0.01)	-13.87 (-2.55)*	25.69 (2.92)* ($H = 5.34$)*
18	15.01 (2.33)	13.99 (2.66)*	-1.69 (-0.26)	-2.51 (-0.55)	-18.04 (-2.39)*	33.05 (3.33)** ($H = 8.04$)**
24	17.63 (2.62)*	14.81 (2.87)*	3.37 (0.45)	-4.13 (-0.76)	-20.25 (-2.98)*	37.88 (3.96)** ($H = 7.46$)**

*Significant at 5% level.

t -Statistics are in parentheses.

**Significant at 1% level.

t -Statistics are in parentheses.

H is the Kruskal–Wallis test statistic. It is approximately distributed as chi-square with 1 degree of freedom. It tests the null hypothesis that the ACAR's of the 'loser' and of the 'winner' portfolios are equal.

$ACAR_{1,t}$ to $ACAR_{5,t}$ series are the average cumulative average market adjusted abnormal returns for the 'loser' portfolio, intermediary portfolios, and 'winner' portfolio, respectively.

$ACAR_{1,t} - ACAR_{5,t}$ series is the difference between 'loser' and 'winner' ACARs.

and

$$t_{5,t} = \frac{ACAR_{5,t}}{S_5/\sqrt{N}}, \quad (6)$$

where S_1 and S_5 are the sample standard deviation of the 'loser' and 'winner' portfolios, respectively.

The test statistic for tests of the third hypothesis is computed by:

$$t_t = \frac{(ACAR_{1,t} - ACAR_{5,t})}{\sqrt{2S_t^2/N}}, \quad (7)$$

where S_t is the standard deviation of the pooled sample of the abnormal returns of 'winner' and 'loser' portfolios of the 8 test periods in month t .

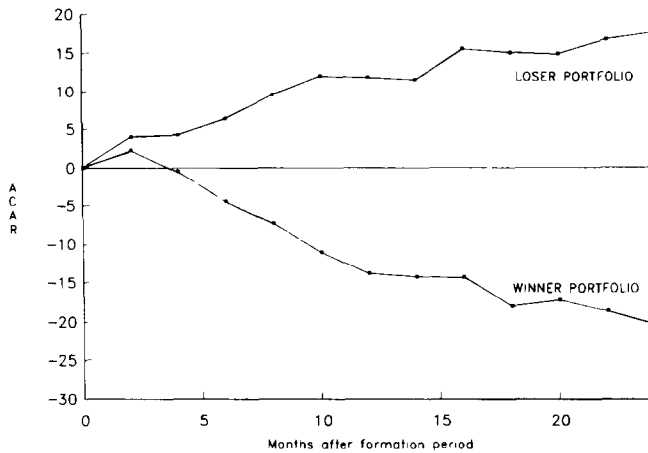


Fig. 1. Average cumulative average abnormal return (ACAR).

4. Empirical results

In Table 1 we report the average cumulative average market adjusted abnormal returns – ACARs for each of the 5 portfolios constructed. The first series ($ACAR_{1,t}$) represents the 'loser' portfolio performance and the fifth series ($ACAR_{5,t}$) the 'winner' portfolio. The results suggest that there is a significant overreaction effect in the Brazilian market, consistent with the U.S. findings of De Bondt and Thaler (1985). After 12 months of the test period the difference in performance between 'loser' and 'winner' portfolios is 25.69% (t -statistic=2.92). After 24 months the 'loser' portfolio outperformed the market by 17.63% (t -statistic=2.62) whereas the 'winner' underperformed the market by 20.25% (t -statistic=-2.98). As in t -tests, normality of the variables is assumed. Table 1 also presents the non-parametric test developed by Kruskal and Wallis to test the null hypothesis that the ACARs of the 'loser' and of the 'winner' portfolios are equal. The results are consistent with those using t -tests. The average performance of the 'winner' and 'loser' portfolios from months 0 to 24 is presented graphically in Fig. 1.

It is informative to look at the intermediary portfolios. For each of the sets of results from month 18 to month 24 of the test period there is a monotonic decrease in abnormal return as one moves from portfolio 1 (the formation period 'loser' portfolio) to portfolio 5 (the formation period 'winner' portfolio).

Table 2

Cumulative average abnormal return for the loser and winner portfolios at the end of each of the 8 formation and test periods (1972/73 to 1988/89)

Formation/test period	Loser portfolio		Winner portfolio		Car loser-car winner
	CAR	CU	CAR	CU	
1972/73	—	—83.94% (-11.64)**	—	99.24% (6.95)**	—
1974/75	23.13% (1.53)	—84.98 (-7.27)**	3.70% (0.22)	82.62 (11.32)**	19.43% (0.95)
1976/77	29.99 (1.21)	—94.06 (-13.51)**	—22.80 (-1.48)	103.83 (10.82)**	52.79 (1.83)
1978/79	—10.72 (-0.68)	—55.24 (-10.68)**	—10.26 (-0.80)	94.29 (11.04)**	—0.46 (-0.02)
1980/81	0.17 (0.06)	—93.99 (-10.72)**	—27.45 (-1.92)	97.04 (14.13)**	27.62 (1.53)
1982/83	5.73 (0.28)	—87.25 (-7.50)**	6.74 (0.48)	101.74 (6.33)**	—1.01 (-0.09)
1984/85	20.86 (1.13)	—109.13 (-13.09)**	—36.30 (-1.90)	116.45 (16.15)**	57.16 (2.12)*
1986/87	50.50 (2.24)*	—117.23 (-15.62)**	—49.05 (-3.16)**	117.43 (13.88)**	99.55 (3.68)**
1988/89	21.41 (0.85)	—	—26.61 (-1.28)	—	48.02 (1.46)

*Significant at 5% level.

**Significant at 1% level.

t-Statistics are in parentheses.

CU is the cumulative average abnormal return at the end of each formation period.

CAR is the cumulative average abnormal return at the end of each test period.

A potential problem in studies of this sort is that the results may be driven by very extreme market behaviour during a relatively small part of the test period. In order to investigate this possibility, we present in Table 2 both the cumulative abnormal return — CU, and the cumulative average abnormal return — CAR for the 'winner' and 'loser' portfolios for each of the 8 formation periods and each of the 8 test periods, respectively. In all of the 8 formation periods the 'winner' and 'loser' CUs are significant at the 1 percent level of confidence. Of the 8 test periods, the 'loser-winner' difference is positive in all but 2 of the sub-periods. Although only three of these positive differences are significantly different from zero, they range in magnitude from 19.43% to 99.55%. There is some evidence that the effect was particularly strong in the period 1984 to 1987, during which time the market rose sharply and then fell sharply. However, there is evidence that the effect existed outside these periods.

4.1. *Can risk differences explain the differences in returns*

One possible explanation for the difference between the returns of the 'loser' and 'winner' portfolios is differential risk. Abnormal returns were computed using the Sharpe–Lintner model. The CAPM expected returns were computed using betas estimated from the monthly returns of each company for the two years before the year in which the returns were generated. The average betas of the securities in the 'winner' and 'loser' portfolios were 1.060 and 1.062 (*t*-statistic for the difference: 0.04).

We followed Chan's (1988) methodology to control for risk differences over time computing betas using test period data as against formation period data.⁵ We regressed the return on an 'arbitrage' portfolio against the market risk premium for each of the 8 two-year test periods, i.e., $R_{L,t} - R_{W,t} = \alpha_A + \beta_A(R_{m,t} - R_{f,t}) + e_{A,t}$, where $R_{L,t}$ and $R_{W,t}$ are the returns on the 'loser' and 'winner' portfolios, respectively; $R_{m,t}$ is the monthly return of the equally-weighted index; and $R_{f,t}$ is the risk-free rate in month t . We find that the average Jensen performance index, $\bar{\alpha} = 0.0150$ with an average *t*-statistic of 3.04. These results show that differences in risk cannot account for the overreaction effect in the Brazilian stock market.

I also calculated the difference between the ACAR's for the 'loser' and 'winner' portfolios using the CAPM-adjusted returns. The 'loser-winner' difference after 24 months was 41.67% (*t*-statistic=2.65). This is broadly consistent with the difference of 37.88% (*t*-statistic=3.96) found for market adjusted returns. Hence, risk-adjustment, at least measured with the Sharpe–Lintner CAPM, is unable to explain the differential performances of the two portfolios.

4.2. *Symmetry versus asymmetry in the overreaction effect*

Our results contrast in one respect with the U.S. results of De Bondt and Thaler (1985) who found that the absolute value of the (positive) average abnormal return from the 'loser' portfolios was much larger than the absolute value of the average abnormal returns from the 'winner' portfolios. Our results suggest that the Brazilian stock market overreaction is symmetric. However, using the 'reversal coefficient' technique developed by Dissanaïke (1992) we found that, although the test-period returns are symmetric, the price reversals relative to formation-period performance are asymmetric. The average 'reversal coefficient' for the 8 two-year test periods was -0.039 (*t*-statistics= -0.21) for the 'loser' portfolios and 0.752 (*t*-

⁵ When calculating the CAPM-beta adjusted returns, data from 1971 and 1972 are used for initial beta estimation.

statistics = 1.87) for the 'winner' portfolios. It means that only the values of the 'winner' portfolios have reverted, relative to the market. This result is similar to the one reported by Dissanaïke (1992) using British data.

5. Conclusion

Using a new set of data for the Brazilian stock market, we find support for the overreaction hypothesis which was documented by De Bondt and Thaler (1985) for U.S. stock returns. The effect seems to have been most pronounced during the period of extreme volatility of the mid-1980s but does not appear to have been confined to that period alone. In contrast to the U.S. evidence, the magnitude of the overreaction effect was more pronounced in Brazil than in the U.S. The results also show that differences in risk, as measured by CAPM beta cannot account for the difference between the 'loser' and 'winner' portfolio returns.

We also examine the issue of asymmetry versus symmetry in the overreaction effect. Using the methodology proposed by Dissanaïke (1992) we find that the overreaction effect is asymmetric. Only the values of the 'winner' portfolios have reverted relative to the market.

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